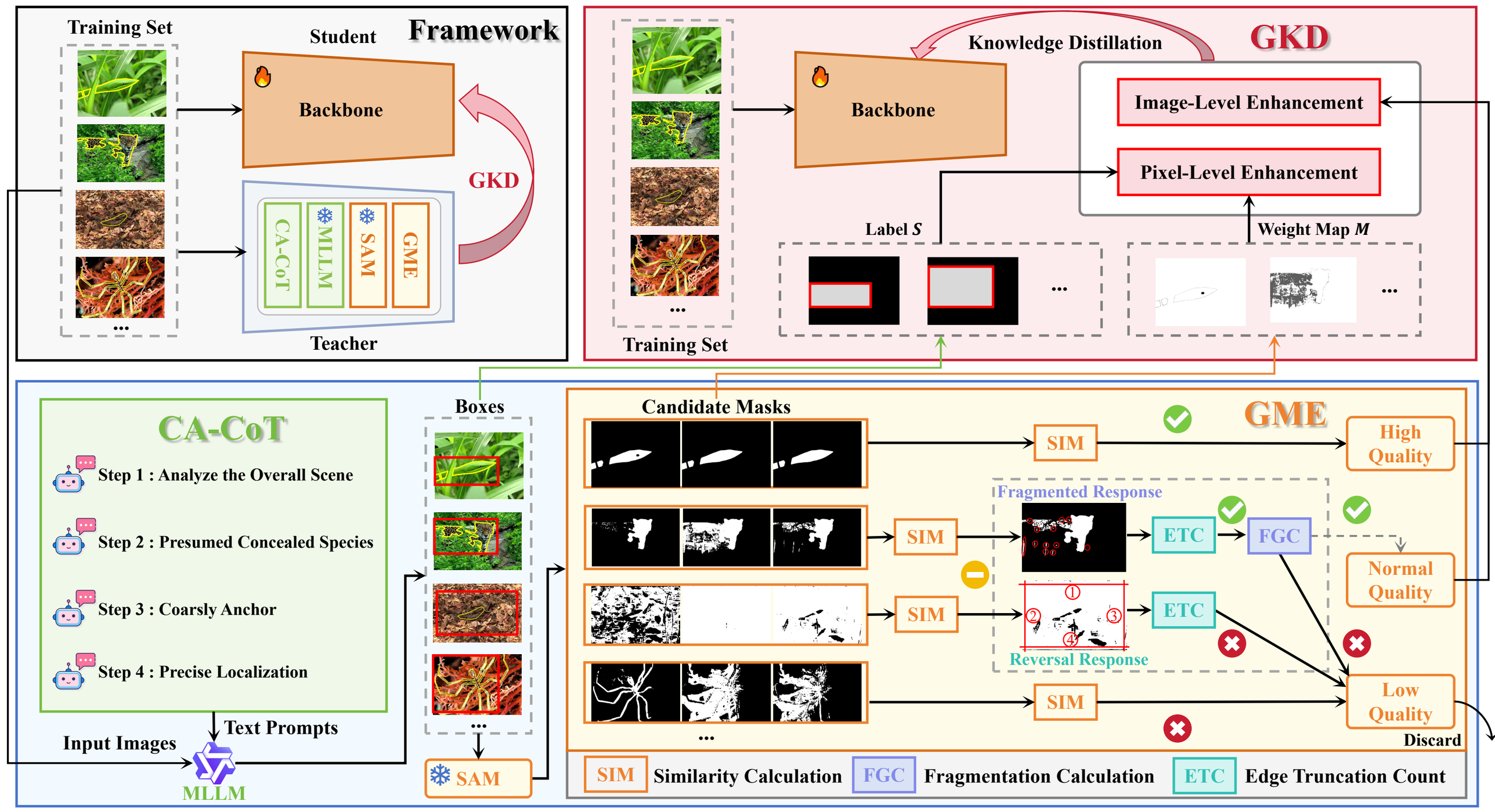
Framework
Training Set
Student
Backbone
GKD
CA-CoT
MLLM
SAM
GME
Teacher
Knowledge Distillation
GKD
Backbone
Image-Level Enhancement
Pixel-Level Enhancement
Label S
Weight Map M
Training Set
CA-CoT
Step 1 : Analyze the Overall Scene
Step 2 : Presumed Concealed Species
Step 3 : Coarsly Anchor
Step 4 : Precise Localization
Text Prompts
Input Images
MLLM
Boxes
SAM
Candidate Masks
GME
SIM
High Quality
Fragmented Response
ETC
FGC
Normal Quality
Reversal Response
Low Quality
Discard
SIM Similarity Calculation
FGC Fragmentation Calculation
ETC Edge Truncation Count